Chapter 1

Supplementary Information for Chapter 2: Helium Step Heat Experiment on Goethite in Hydrothermal Conditions

1.1 Goethite Permeation Rate Constant Dataset

The in-experiment permeation rate constants used by the permeation step method below – and the isothermal-permeation-fit methodology, cooling-tube numerical model, and Arrhenius regression used to determine them – are described in full in the main text (§**??**). The complete fitted dataset underlying that regression is reported here: Table 1.1 gives the raw per-step data for all 10 isothermal sequences, and Table 1.2 gives the lambda/n_0 fit to each sequence (Fig. **??**) that together define the Arrhenius regression of Fig. **??**.

Table 1.1: Raw step-heat data ('Fit Lambdas' sheet, cross-checked against 'Raw Data and Corrections') for the 10 isothermal 3He permeation sequences used to determine the ampoule's apparent permeability (Fig. **??**, Fig. **??**). Series are separated by horizontal rules; within a series, steps are listed in the order measured.

Series	Temp (°C)	Nominal Time (min)	Cooling Time (min)	^{3}He measured (pccSTP)	$\pm 2\sigma$
1	105.00	150	26	0.0190	1.1×10^{-4}
1	105.00	150	26	0.0111	7.4×10^{-5}
1	105.00	150	26	0.0080	5.3×10^{-5}
2	109.99	150	26	0.0229	1.4×10^{-4}
2	110.00	150	26	0.0115	7.3×10^{-5}
2	110.00	150	26	0.0075	5.2×10^{-5}
3	99.86	120	26	0.0106	6.1×10^{-5}
3	100.00	120	26	0.0074	4.0×10^{-5}
3	100.00	120	26	0.0049	2.5×10^{-5}
3	99.99	120	26	0.0043	2.1×10^{-5}
4	95.00	150	26	0.0109	2.8×10^{-4}
4	95.00	150	26	0.0081	2.0×10^{-4}

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Series	Temp (°C)	Nominal Time (min)	Cooling Time (min)	^{3}He measured (pccSTP)	$\pm 2\sigma$
4	95.00	150	26	0.0062	1.4×10^{-4}
5	104.99	120	26	0.0215	8.9×10^{-5}
5	104.99	120	26	0.0144	6.0×10^{-5}
5	104.99	120	26	0.0104	4.3×10^{-5}
6	100.00	120	26	0.0126	5.6×10^{-5}
6	100.00	120	26	0.0094	3.5×10^{-5}
6	100.00	120	26	0.0074	2.0×10^{-5}
7	95.00	120	26	0.0072	3.0×10^{-5}
7	95.00	120	26	0.0063	2.6×10^{-5}
7	95.00	120	26	0.0045	1.9×10^{-5}
8	90.00	150	26	0.0059	2.7×10^{-5}
8	89.90	150	26	0.0048	2.4×10^{-5}
8	90.00	150	26	0.0036	1.9×10^{-5}
9	90.00	180	26	0.0100	6.2×10^{-5}
9	89.92	180	26	0.0067	5.1×10^{-5}
9	90.00	180	26	0.0054	4.8×10^{-5}
10	99.99	150	26	0.0104	4.3×10^{-5}
10	99.98	150	26	0.0068	3.5×10^{-5}
10	100.00	150	26	0.0054	3.4×10^{-5}

Table 1.2: Fitted λ (permeation rate constant) and n_0 (residual helium at the start of the series) for each of the 10 isothermal sequences in Table 1.1, from the weighted non-linear least-squares fit of Eq. (**??**) (Chapter **??**) described in the main text (§**??**). n is the number of steps in the sequence. These are the same per-sequence fits plotted in Fig. **??** and used to build the Arrhenius regression in Fig. **??**.

Series	Temp (°C)	Duration (min)	n	λ (s^{-1})	$\pm 2\sigma$	n_0 (pccSTP)	$\pm 2\sigma$
1	105.00	150	3	4.9×10^{-5}	1.4×10^{-5}	0.0521	7.8×10^{-3}
2	109.99	150	3	6.3×10^{-5}	1.8×10^{-5}	0.0508	7.3×10^{-3}
3	99.86	120	4	4.3×10^{-5}	1.3×10^{-5}	0.0377	5.2×10^{-3}
4	95.00	150	3	3.1×10^{-5}	2.6×10^{-6}	0.0444	2.2×10^{-3}
5	104.99	120	3	5.0×10^{-5}	6.2×10^{-6}	0.0696	4.7×10^{-3}
6	100.00	120	3	3.7×10^{-5}	4.1×10^{-6}	0.0534	3.4×10^{-3}
7	95.00	120	3	3.3×10^{-5}	1.5×10^{-5}	0.0348	1.1×10^{-2}
8	90.00	150	3	2.7×10^{-5}	4.8×10^{-6}	0.0279	3.5×10^{-3}
9	90.00	180	3	3.0×10^{-5}	9.0×10^{-6}	0.0357	7.0×10^{-3}
10	99.99	150	3	3.8×10^{-5}	1.2×10^{-5}	0.0351	7.0×10^{-3}

Permeation Step Method Step-Heat Dataset

The complete corrected step-heat dataset produced by the permeation step method, using the in-experiment rate constants above (Fig. **??**), is reported in Table 1.3, giving the central value and 2σ uncertainty for each quantity, analogous to Table **??** in the main text for the bounding method. 4He is not treated by the permeation step method here: 4He permeation-step signals were blank level, so no direct, step-by-step quantification of the ampoule's 4He inventory was possible, and 4He throughout this chapter is instead reduced exclusively by the bounding method of the main text (§**??**). The permeation step dataset reported here is retained only for 3He, as the internal check on the bounding method discussed in §**??**.

Table 1.3: ^{3}He step-heat dataset for HT-Win-06-01b under the permeation step method: measured and cumulative ^{3}He, the initial and residual ampoule content (n_0, n_t), the true step production (ω), and the apparent diffusivity $\ln(D/a^2)$, each with its 2σ uncertainty. Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal, as in Table **??**.

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	n_0 (pccSTP)	$\pm 2\sigma$	n_t (pccSTP)	$\pm 2\sigma$	ω (pccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
170	300	0.9630	0.0000	0.00×10^{0}	0.00×10^{0}	0.1968	2.22×10^{-2}	1.1597	2.63×10^{-2}	-21.910	0.045
180	463	0.5442	0.9630	7.26×10^{-2}	1.29×10^{-2}	0.0782	4.93×10^{-3}	0.5497	1.60×10^{-2}	-22.184	0.089
190	550	0.5292	1.6241	1.95×10^{-2}	2.54×10^{-3}	0.0679	8.20×10^{-3}	0.5777	1.30×10^{-2}	-21.976	0.085
200	500	0.5190	2.2112	3.28×10^{-2}	5.42×10^{-3}	0.0532	5.77×10^{-3}	0.5394	1.15×10^{-2}	-21.702	0.088
210	475	0.5770	2.7646	1.41×10^{-2}	2.11×10^{-3}	0.0552	5.71×10^{-3}	0.6182	1.17×10^{-2}	-21.311	0.082
220	450	0.6333	3.3799	1.10×10^{-2}	1.80×10^{-3}	0.0469	4.58×10^{-3}	0.6692	1.24×10^{-2}	-20.991	0.080
230	425	0.7899	4.0551	5.87×10^{-3}	1.86×10^{-3}	0.0639	3.48×10^{-3}	0.8479	1.25×10^{-2}	-20.514	0.066
225	908	0.7513	4.8885	1.95×10^{-2}	2.45×10^{-3}	0.0420	4.94×10^{-3}	0.7738	1.42×10^{-2}	-21.200	0.121
215	450	0.1415	5.6842	1.31×10^{-2}	1.85×10^{-3}	0.0278	3.54×10^{-3}	0.1563	6.48×10^{-3}	-22.015	0.316
205	520	0.0763	5.8529	$7.18\times10^{-3*}$	$1.23\times10^{-3*}$	0.0125*	$1.63\times10^{-3*}$	0.0816*	$8.10\times10^{-3*}$	-22.788*	0.829*
195	1106	0.0670	5.9491	$2.12\times10^{-3*}$	$4.02\times10^{-4*}$	0.0307*	$2.86\times10^{-3*}$	0.0956*	$3.23\times10^{-3*}$	-23.370*	1.094*
185	740	0.0227	6.0246	$1.25\times10^{-2*}$	$1.64\times10^{-3*}$	0.0090*	$1.06\times10^{-3*}$	0.0192*	$2.00\times10^{-3*}$	-24.567*	2.822*
175	920	0.0180	6.0610	$3.68\times10^{-3*}$	$5.02\times10^{-4*}$	0.0045*	$4.42\times10^{-4*}$	0.0189*	$7.42\times10^{-4*}$	-24.796*	3.797*
200	260	0.0224	6.0814	$6.53\times10^{-4*}$	$1.17\times10^{-4*}$	0.0095*	$1.23\times10^{-3*}$	0.0313*	$1.65\times10^{-3*}$	-23.022*	1.495*
210	520	0.0812	6.1081	$7.48\times10^{-3*}$	$9.98\times10^{-4*}$	0.0145*	$1.32\times10^{-3*}$	0.0882*	$3.85\times10^{-3*}$	-22.669*	1.135*
220	1167	0.5766	6.1914	2.82×10^{-3}	1.44×10^{-3}	0.0276	3.24×10^{-3}	0.6014	2.45×10^{-2}	-21.504	0.184
230	370	0.3577	6.7796	8.68×10^{-3}	3.96×10^{-3}	0.0422	5.13×10^{-3}	0.3912	1.48×10^{-2}	-20.712	0.146
240	320	0.6767	7.1563	1.72×10^{-2}	2.42×10^{-3}	0.0826	9.94×10^{-3}	0.7420	1.57×10^{-2}	-19.807	0.099
245	250	0.5549	7.8582	2.13×10^{-2}	8.87×10^{-3}	0.0516	5.65×10^{-3}	0.5852	1.46×10^{-2}	-19.710	0.125
250	210	0.6934	8.4743	1.08×10^{-2}	4.52×10^{-3}	0.0718	8.15×10^{-3}	0.7545	1.41×10^{-2}	-19.200	0.102
245	270	0.4659	9.2086	2.48×10^{-2}	3.48×10^{-3}	0.0498	5.84×10^{-3}	0.4909	1.01×10^{-2}	-19.810	0.197
240	320	0.2857	9.7206	2.07×10^{-2}	2.81×10^{-3}	0.0339	4.08×10^{-3}	0.2989	6.58×10^{-3}	-20.433	0.312
235	370	0.2199	10.0357	1.66×10^{-2}	2.20×10^{-3}	0.0273	3.29×10^{-3}	0.2307	5.23×10^{-3}	-20.810	0.401
230	420	0.1400	10.2736	1.32×10^{-2}	1.76×10^{-3}	0.0255	2.53×10^{-3}	0.1523	4.00×10^{-3}	-21.332	0.603
225	652	0.1494	10.4279	1.26×10^{-2}	1.62×10^{-3}	0.0131	1.47×10^{-3}	0.1499	3.42×10^{-3}	-21.772	0.747
230	500	0.1550	10.5908	7.34×10^{-4}	1.67×10^{-3}	0.0128	1.32×10^{-3}	0.1671	3.37×10^{-3}	-21.383	0.461
235	450	0.1951	10.7582	2.57×10^{-3}	4.18×10^{-4}	0.0334	1.85×10^{-3}	0.2259	3.29×10^{-3}	-20.956	0.364
240	400	0.2425	10.9638	6.90×10^{-3}	1.45×10^{-3}	0.0251	3.16×10^{-3}	0.2607	1.23×10^{-2}	-20.672	0.318
245	350	0.3150	11.2328	5.79×10^{-3}	1.05×10^{-3}	0.0395	4.64×10^{-3}	0.3487	9.39×10^{-3}	-20.219	0.240
250	300	0.3716	11.5689	1.65×10^{-2}	2.24×10^{-3}	0.0467	5.24×10^{-3}	0.4018	9.62×10^{-3}	-19.888	0.209
250	300	0.3640	11.9627	2.43×10^{-2}	4.37×10^{-3}	0.0339	3.82×10^{-3}	0.3735	7.99×10^{-3}	-19.927	0.242

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Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	n_0 (pccSTP)	$\pm 2\sigma$	n_t (pccSTP)	$\pm 2\sigma$	ω (pccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
250	300	0.3452	12.3490	1.13×10^{-2}	1.60×10^{-3}	0.0395	4.48×10^{-3}	0.3733	7.91×10^{-3}	-19.895	0.244
250	10250	8.4829	12.7167	1.91×10^{-2}	2.53×10^{-3}	0.1165	7.38×10^{-3}	8.5804	1.28×10^{-1}	-19.958	0.189
240	785	0.2261	21.2187	2.90×10^{-9}	6.85×10^{-9}	0.0263	2.92×10^{-3}	0.2525	2.21×10^{-2}	-20.660	0.580
Total			146.1483								

1.1.1 Permeation Step Approach and Concordance with Bounding Approach

Beyond the ampoule's residual helium content itself (§**??**), the two methods' derived apparent diffusivities also agree closely. Fig. 1.1 compares $\ln(D/a^2)$ from the permeation-step method against the bounding method's central value for every step in the experiment, with the bounding method's own lower- and upper-bound spread on one axis and the permeation step method's propagated uncertainty on the other. The two diffusivity datasets track the 1:1 line within uncertainty across the full range of the experiment. Fitting the permeation-step method's own diffusivities with the same Stage 1 procedure used for the bounding method in the main text (Fig. 1.2) yields an activation energy of 157.0±4.8 kJ/mol (R^2=0.991, n=12) and a closure temperature of 97.5±7.8°C, indistinguishable from the bounding method's own 157.1±5.3 kJ/mol and 98.1±8.6°C (Fig. **??**, main text).

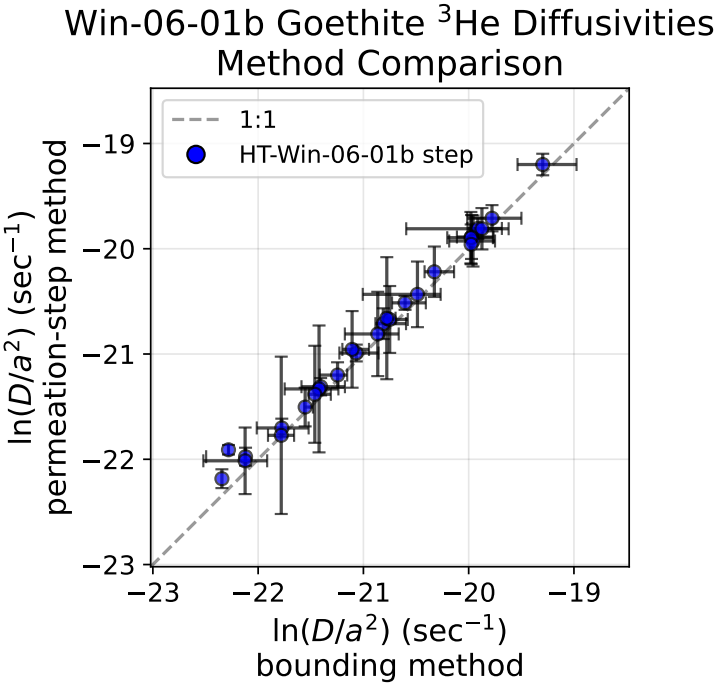


Figure 1.1: Apparent ^{3}He diffusivities for HT-Win-06-01b from the permeation-step method (y-axis) against the bounding method's central value (x-axis, Eq. (**??**)), with the bounding method's lower/upper-bound spread and the permeation step method's own propagated uncertainty plotted on their respective axes. The dashed line is 1:1.

1.2 Kinetic mechanisms and the limits of a single heating schedule

Solid-state rate laws take the form $d\alpha/dt = A\exp(-E_a/RT)\,h(\alpha)$. For any heating schedule this integrates to

$$g(\alpha) \equiv \int_0^\alpha \frac{d\alpha'}{h(\alpha')} = x, \qquad x = A\sum_j \Delta t_j \exp\left(-\frac{E_a}{RT_j}\right). \tag{1.1}$$

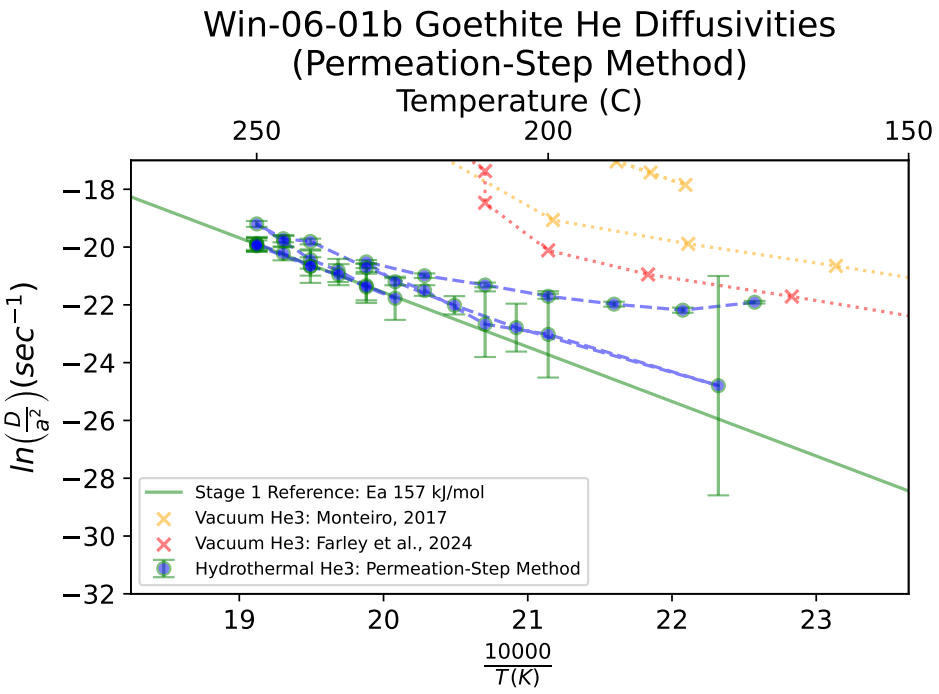


Figure 1.2: Analogue of Fig. **??** (main text) using the permeation-step method's own ^{3}He diffusivities and its own Stage 1 linear regression, rather than the bounding method's. The resulting activation energy and closure temperature (157.0±4.8 kJ/mol; 97.5±7.8°C) agree closely with the bounding method's own (157.1±5.3 kJ/mol; 98.1±8.6°C).

We tested the mechanisms in Table 1.4, including spherical diffusion on the same footing. Under vacuum, the driving-force factor $1-\exp(-\Delta\mu/RT)$ is unity to better than 10^{-9}, so x is simply an Arrhenius-weighted time. Diffusion and conversion then differ only in the shape $\alpha(x)$ and in the fitted E_a.

Table 1.4: Kinetic models and their integrated forms.

Model	$h(\alpha)$	$g(\alpha)$	$\alpha(x)$
R2 contracting cylinder	$2(1-\alpha)^{1/2}$	$1-(1-\alpha)^{1/2}$	$1-(1-x)^2$
R3 contracting sphere	$3(1-\alpha)^{2/3}$	$1-(1-\alpha)^{1/3}$	$1-(1-x)^3$
F1 first order	$1-\alpha$	$-\ln(1-\alpha)$	$1-e^{-x}$
A2, A3 Avrami ($n=2,3$)	$n(1-\alpha)[-\ln(1-\alpha)]^{1-1/n}$	$[-\ln(1-\alpha)]^{1/n}$	$1-e^{-x^n}$
D3 sphere diffusion	—	—	$1-\dfrac{6}{\pi^2}\displaystyle\sum_n \dfrac{e^{-n^2\pi^2 x}}{n^2}$

Synthetic test. We generated noiseless release curves from the R3 model ($E_a = 150$ kJ mol^{-1}) on two schedules: a monotonic series of 1 h steps from 150 to 390 °C, and a cycled schedule with retrograde steps and long isothermal holds. We then fitted every model with E_a and A free. Table 1.5 shows that every model except the Avrami forms fits within 1% in F by adjusting E_a. With realistic analytical uncertainty, goodness of fit to a single schedule cannot identify the mechanism. The fitted activation energy trades off against the shape of $\alpha(x)$. Discrimination requires a second, independent schedule, an independent measure of α, or both. The vacuum experiments supply both (§**??**).

Table 1.5: Fits to synthetic R3 data (E_a in kJ mol^{-1}; rms misfit in F).

	Monotonic schedule		Cycled schedule	
Model	E_a	rms	E_a	rms
R2	140.0	0.0031	140.4	0.0028
F1	173.0	0.0074	177.4	0.0092
A2	81.7	0.0073	83.8	0.0077
A3	51.3	0.0073	53.8	0.0062
D3	296.2	0.0026	296.9	0.0025
R3 (true)	150.0	0.0000	150.0	0.0000

Fits to the vacuum data. Table 1.6 gives the fit of each model to the cumulative KAF ^{3}He release within the fitting window. It also gives the misfit of the resulting out-of-sample prediction of HDM, and the fit to the KAF water data. F1 gives the best fit and the best prediction. R3 and R2 are close behind, so we describe the mechanism as nucleation- or interface-controlled rather than claim first order specifically. The Avrami fits reach the lower bound of E_a and fail on both data sets. Spherical diffusion fits KAF five times worse than F1 and cannot produce the $F \approx 0.89$ plateau.

The fitting window excludes the low-temperature steps, where release is dominated by diffusion from goethite, and the plateau, where it is dominated by diffusion from hematite; the rms misfits are computed over the same steps. Repeating the fits with every step included gives the same result: F1 remains the best model ($E_a = 122.7$ kJ mol^{-1}, $\phi = 0.112$, rms 0.0073 for KAF and 0.026

for the HDM prediction), followed by R3 (99.4 kJ mol^{-1}; 0.0097, 0.030) and R2 (87.9 kJ mol^{-1}; 0.0116, 0.033), with D3 at 175.9 kJ mol^{-1} (0.063, 0.061).

Table 1.6: Model fits to the vacuum experiments (from `vacuum_kinetics_KAF_to_HM_fits.csv`). s: rate factor for the trap-isolated steps.

Model	E_a	$\ln A$	ϕ	s	rms KAF	rms HDM	E_a (water)
R2	90.3	10.00	0.132	0.54	0.0133	0.0292	64.7
R3	101.7	12.38	0.127	0.50	0.0111	0.0234	73.4
F1	122.9	18.67	0.112	0.46	0.0085	0.0167	91.5
A2	25.6	−4.70	0.000	0.20	0.0207	0.1814	20.2
A3	20.0^a	−5.89	0.159	0.00	0.0446	0.2348	20.0^a
D3	183.7	29.50	—	1.70	0.0455	0.0529	180.2

E_a in kJ mol^{-1}, A in s^{-1}. ^{a}At the lower bound of the fit.

1.3 Sensitivity of the size effect to the surface-area basis

The sign of the surface stabilization depends on how the surface areas of the two phases are related. At equal area per mole of Fe (Eq. (**??**)), goethite is stabilized whenever $\gamma_{\mathrm{hem}} > \gamma_{\mathrm{goe}}$. If both phases are instead spheres of the same diameter d, then $A_i = 6V_i/d$ and

$$\Delta G_{\mathrm{surf}} = \frac{6}{d}\left(\gamma_{\mathrm{hem}} V_{\mathrm{hem}} - 2\gamma_{\mathrm{goe}} V_{\mathrm{goe}}\right), \tag{1.2}$$

which stabilizes goethite only if

$$\frac{\gamma_{\mathrm{hem}}}{\gamma_{\mathrm{goe}}} > \frac{2V_{\mathrm{goe}}}{V_{\mathrm{hem}}} = \frac{2 \times 20.881}{30.274} = 1.379. \tag{1.3}$$

Hematite is denser and carries less surface per Fe at equal size. For hydrated surfaces, $\gamma_{\mathrm{hem}}/\gamma_{\mathrm{goe}} = 0.75/0.60 = 1.25 < 1.379$, so fine goethite is destabilized: at 10 nm, $\Delta G_{\mathrm{surf}} = (6/10^{-8})(0.75 \times 30.274 - 2 \times 0.60 \times 20.881) \times 10^{-6} = -1.41$ kJ. For anhydrous surfaces, $1.90/0.91 = 2.09 > 1.379$ and $\Delta G_{\mathrm{surf}} = +11.7$ kJ at 10 nm. Table 1.7 compares the two bases. The equal-area basis corresponds to pseudomorphic conversion and underlies the main-text calculations. The equal-diameter basis is more appropriate where hematite grows as separate crystals, as in fluid-mediated conversion (§**??**). There the surface term slightly favors hematite, but it is negligible at micrometer sizes.

Table 1.7: ΔG_{surf} (kJ per reaction as written) for the two surface-area bases.

	Hydrated		Anhydrous	
d	equal area	equal diameter	equal area	equal diameter
1 μm	+0.04	−0.01	+0.25	+0.12
100 nm	+0.38	−0.14	+2.48	+1.17
10 nm	+3.76	−1.41	+24.8	+11.7

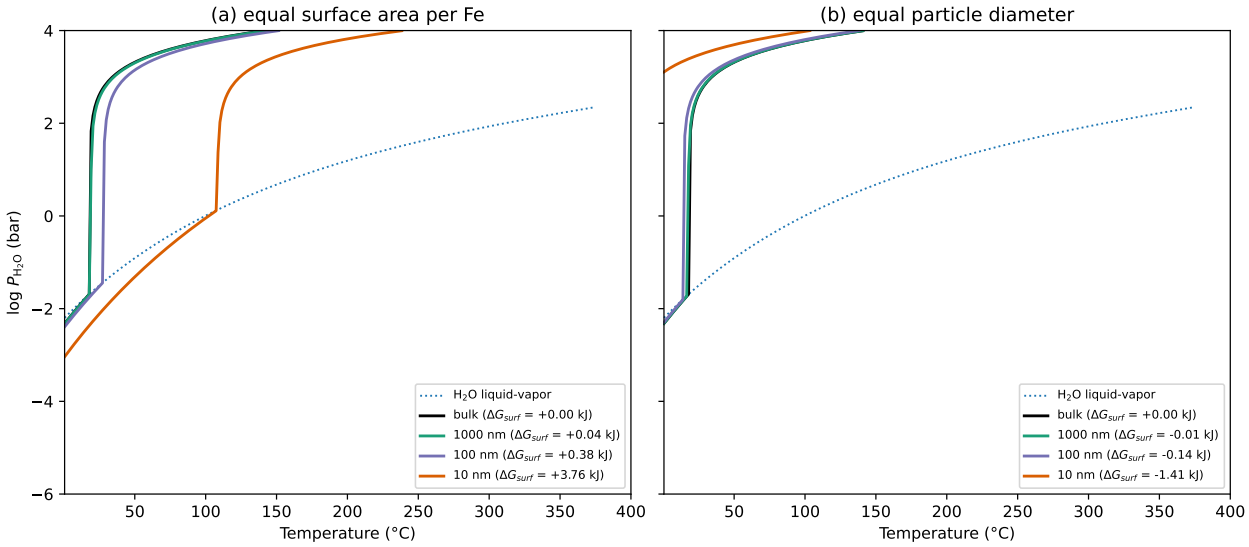


Figure 1.3: Goethite–hematite–water boundaries for hydrated surfaces at (a) equal surface area per mole of Fe and (b) equal particle diameter. The size effect changes sign between the two bases.

1.4 Nucleation barrier ratio and the calibration of γ_{eff}

From Eq. (**??**), the ratio of barriers at water saturation and under vacuum at the same temperature is

$$\frac{\Delta G^*_{\mathrm{sat}}}{\Delta G^*_{\mathrm{vac}}} = \left(\frac{\Delta\mu_{\mathrm{vac}}}{\Delta\mu_{\mathrm{sat}}}\right)^2 \left(\frac{\gamma_{\mathrm{sat}}}{\gamma_{\mathrm{vac}}}\right)^3 . \tag{1.4}$$

The prefactors, the temperature and the molar volume cancel. The first factor follows from the thermodynamic data alone (Table **??**; 99 at 250 °C for bulk goethite). The second is 1 if nucleation occurs at the same internal interface in both settings. It is $(0.75/1.90)^3 = 1/16$ in the extreme case that the relevant interface is a free hematite surface, hydrated at saturation and anhydrous under vacuum. Figure 1.4 shows the ratio between these bounds for bulk and 10 nm goethite. Even at the lower bound, the saturated barrier is about six times the vacuum barrier at 250 °C. Because the barrier appears in an exponent, that is still a difference of many orders of magnitude in rate.

Figure 1.5a shows γ_{eff} from Eq. (**??**) as a function of the assumed water pressure during the trap-isolated steps. It spans 0.08–0.20 $\mathrm{J\,m^{-2}}$ over 10^{-5}–10^{-1} bar and 0.14–0.19 $\mathrm{J\,m^{-2}}$ over the volume-constrained range (Table **??**). Figure 1.5b shows the predicted slowdown at saturation (Eq. (**??**)) for the calibrated values and for the most permissive value.

Sensitivity to surface terms. The calibration and translation use bulk driving forces (§**??**). To test this choice, we repeat both with the surface term of Eq. (**??**) included in $\Delta\mu$. We use anhydrous surfaces for the vacuum and trap-isolated steps and hydrated surfaces at saturation, the combination that most reduces the vacuum driving force relative to the saturated one. Table 1.8 shows that the conclusion is unchanged down to 10 nm grains.

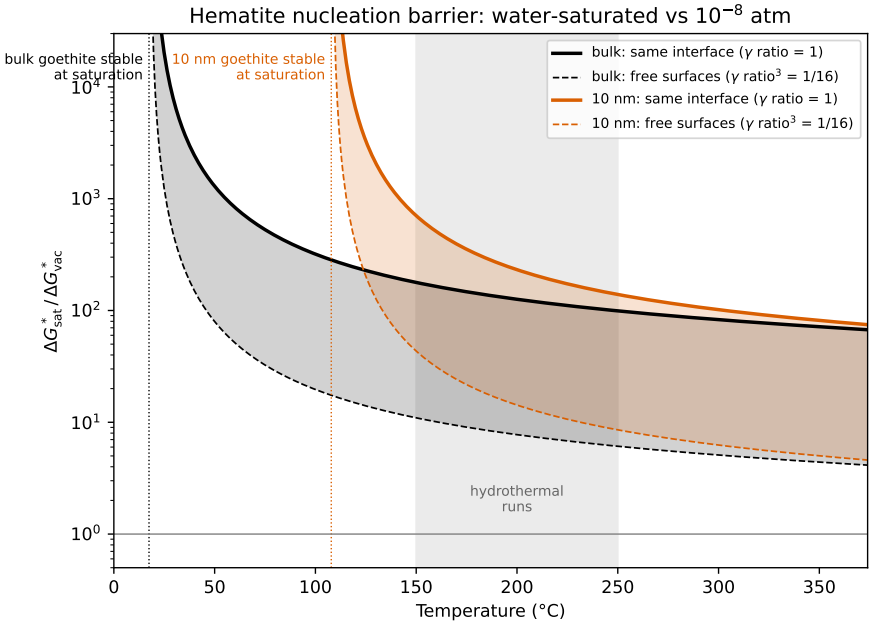


Figure 1.4: Ratio of hematite nucleation barriers at water saturation and at $P_{H_2O} = 10^{-8}$ atm for bulk and 10 nm goethite. Solid: same interface in both settings; dashed: free hematite surfaces (hydrated vs anhydrous). Below the vertical dotted lines, goethite is stable at saturation and there is no driving force.

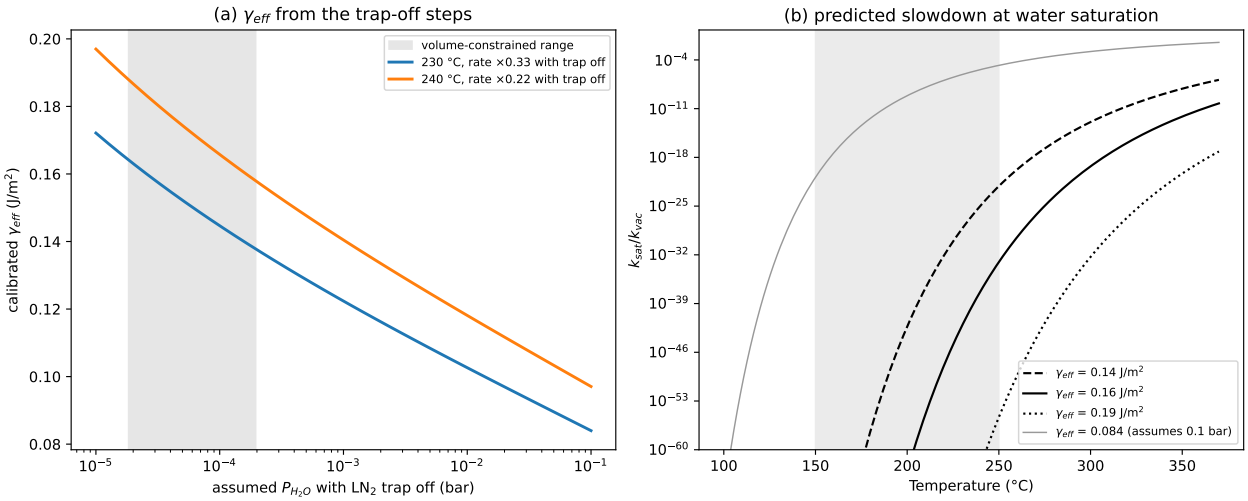


Figure 1.5: (a) γ_{eff} calibrated from the two trap-isolated KAF steps as a function of the assumed P_{H_2O}; grey band: range allowed by the apparatus volume (190–600 mL) and the water released in each step. (b) Ratio of conversion rate constants at water saturation and under vacuum (Eq. (**??**)); grey band: temperatures of the hydrothermal experiments.

Table 1.8: Effect of including surface terms in $\Delta\mu$ on the calibration and the predicted hydrothermal conversion. Driving forces in kJ per mole Fe_2O_3 at 250 °C; γ_{eff} range over Table **??**.

Grain size	$\Delta\mu_{\mathrm{vac}}$	$\Delta\mu_{\mathrm{sat}}$	γ_{eff} (J m^{-2})	$\log_{10}\alpha_{\mathrm{HT}}$
Bulk	106.0	10.65	0.138–0.188	−32.0
100 nm	103.5	10.27	0.133–0.182	−31.2
10 nm	81.2	6.89	0.090–0.131	−23.3

α_{HT}: predicted conversion over the hydrothermal schedule, median γ_{eff}.